Network modeling – AY 2026/2027
Homework 1 – Modeling Zoom video uplink traffic

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Abstract

I analyze one outbound UDP flow from a Zoom meeting between a PC and an iPad. Packets occur in tight clusters, which a Poisson source fails to reproduce. An empirical batch-renewal model captures packet inter-arrival and size distributions and matches the mean rate on a held-out segment. It does not reproduce the observed correlation of traffic over longer time scales.

1 Measurement and sample selection

I captured Zoom traffic with Wireshark on the PC's Ethernet interface `enp5s0`; the iPad was a second participant on the same LAN. The experiment progressed through idle/muted operation, a connected meeting with media off, microphone-only speech and silence, camera-on speech, and camera off again. These phase durations were approximate: the stopwatch ran for about ten minutes, so I do not treat the planned durations as exact boundaries.

The video-related flow was identified by switching the PC camera off and on and observing its traffic decrease and recover. Its endpoints were `192.168.3.7:42613` and `206.247.58.75:8801`. I selected only the PC-to-server direction using

```
udp.stream == 0 && ip.src == 192.168.3.7 && frame.number > 32864.
```

The filter retains 34,319 packets spanning 615.08 s. I set $t = 0$ at the first selected packet (431.2223 s into the raw capture), not at the stopwatch start. The rate rises near $t = 260$ s and falls near $t = 578$ s (Fig. 1). I therefore select the interior camera-on interval $[270, 570)$ s: 32,136 packets, 300 s, and 862.50 kbit/s. I fit the model on the first 180 s and reserve the last 120 s for validation. No selected packet is truncated. Packet sizes are `frame.len` in bytes, including captured Ethernet headers, not encoded media payload.

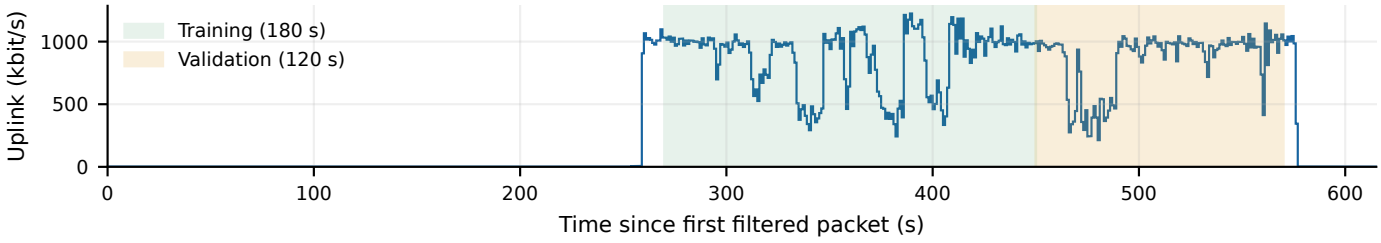


Figure 1: One-second uplink rate and the selected training and validation intervals. Camera transitions are inferred from traffic; audio-only phase boundaries are not resolved in this flow.

2 Source model

For packet timestamps t_j, define packet inter-arrival times as $A_j^{\text{pkt}} = t_j - t_{j-1}$. In the training segment, these have mean 9.162 ms, median 5.38 μs, and coefficient of variation 1.711; 63.17% are at most 1 ms. I first fit a Poisson baseline: $A_j^{\text{pkt}} \sim \text{Exp}(109.145\,\text{s}^{-1})$, one packet per event, and independent packet sizes sampled from the training empirical distribution. Its exponential inter-arrivals cannot represent the concentration near zero.

Following Lessons 3–4, I model the bursty VBR source with a marked and weighted renewal process, augmented with packet offsets. Consecutive packets separated by at most $\delta = 1$ ms belong to the same cluster. These clusters are defined by packet timing; they do not identify codec frames. Let τ_i be a cluster's first timestamp, B_i its packet count, $U_{i,k}$ its packet offsets, and $W_{i,k} = 8L_{i,k}$ its weights in bits. Each complete training cycle stores the joint tuple

$$C_i = (D_i, B_i, \{U_{i,k}, W_{i,k}\}_{k=1}^{B_i}), \qquad D_i = \tau_{i+1} - \tau_i.$$

I exclude the boundary clusters, leaving 7,233 cycles. To generate a trace, I draw cycles independently and uniformly from this empirical distribution, emit packets at $\tau_i + U_{i,k}$, then advance by D_i. Here D_i is the inter-arrival time between cluster starts, distinct from the packet inter-arrival time A_j^{pkt} plotted in Fig. 2. Weights, counts and offsets remain dependent within a sampled cycle. Offsets prevent artificial zero packet inter-arrivals. The fitted means are $E[D] = 24.881$ ms and $E[B] = 2.716$ packets, with predicted rate $E[\sum_k W_{i,k}]/E[D] = 865.95$ kbit/s. Thresholds of 0.5 and 3 ms give mean batch sizes 2.707 and 2.774, close to the value at 1 ms.

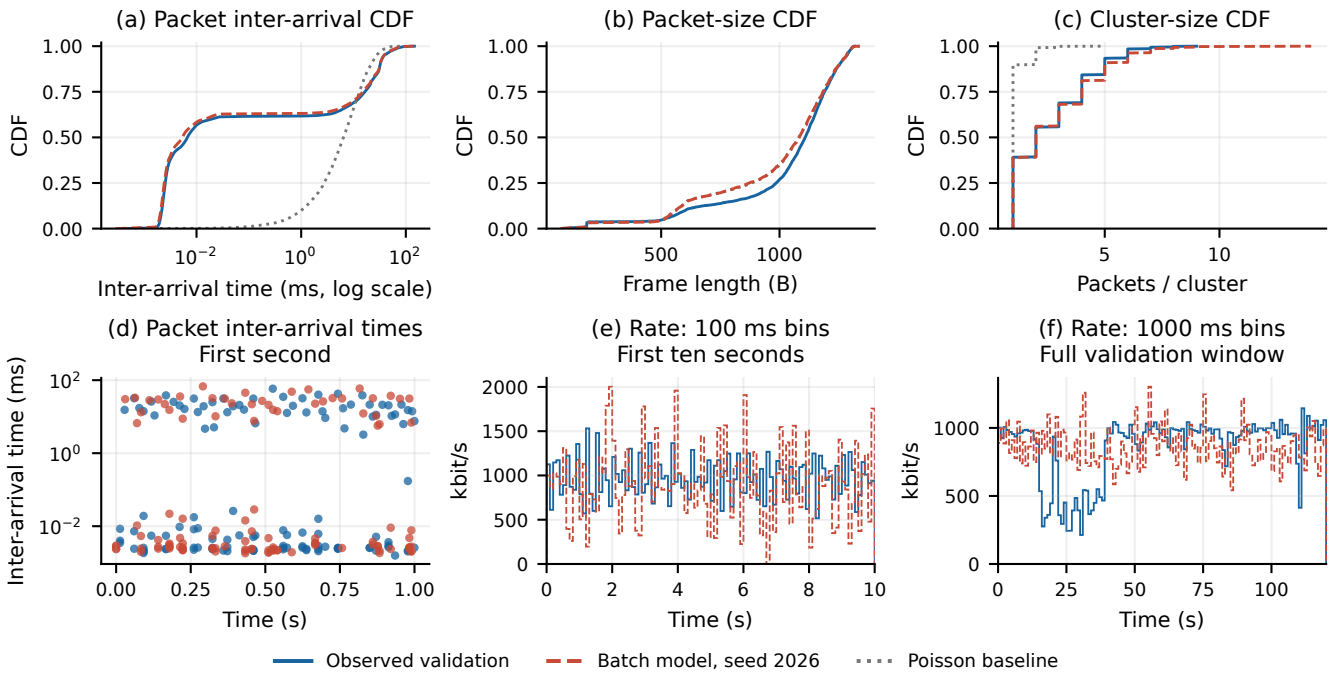


Figure 2: Held-out observations and independent 120 s simulations (seed 2026). CDFs (a–c) use the full traces; (a) and scatterplot (d) show packet inter-arrival times A_j^{pkt}, not cluster intervals D_i. The scatterplot shows the first second. Cluster sizes in (c) use the same 1 ms rule for every trace. Rates use 100 ms bins over the first ten seconds (e) and 1 s bins over all 120 s (f). Synthetic events are not expected to align with observed events.

3 Results

120 s validation comparison	Observed	Poisson	Batch model
Packet count	12,491	12,927	13,102
Mean frame length (B)	1,029.57	989.97	990.38
Mean rate (kbit/s)	857.35	853.16	865.06
Packet inter-arrival CDF distance	—	0.611	0.050
Count Fano factor, 100 ms bins	1.115	0.995	2.475
Lag-1 count correlation, 1 s bins	0.684	0.151	0.049

Both simulations use training data only and stop at 120 s; packet counts are stochastic, not forced to match observations. The batch simulation's mean rate is 0.90% above validation. Across 30 seeds (2026–2055), its mean rate is 864.30 kbit/s; the empirical 5th–95th percentile range is 847.42–879.97 kbit/s. Figure 2(a) shows that the batch model preserves short packet inter-arrivals much better than Poisson: the maximum absolute empirical CDF difference falls from 0.611 to 0.050. This is a descriptive distance, not an independence-based significance test. The size-CDF distance is 0.089 for the batch model, reflecting some shift in sizes between training and validation. The full-window rate comparison in Fig. 2(f) exposes slower variations that are absent from the independently sampled cycles.

4 Discussion

The batch model captures local clustering and average load, but overestimates 100 ms count variability and loses the strong correlation between successive one-second counts. Independent cycle sampling discards ordering across clusters; it therefore misses slow activity or adaptation changes. A state-dependent or block model could address this limitation. This single run covers one video-related UDP flow, not all Zoom traffic or separately identified audio. Phase boundaries are inferred and payloads are encrypted. The model describes packets observed at the host capture interface, not the encoder's internal generation times or a universal Zoom traffic law. Controlled camera changes alone do not provide enough cycles to estimate an On–Off sojourn-time model. Course basis: Lesson 3, *Network abstraction and traffic source models*; Lesson 4, *Traffic source models*.